UNIVERSITY OF RWANDA
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Applying Human-Centered Design

File Management AI Assistant

Course: Human-Centered Design

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1. Introduction

Students often keep their course files in many places such as phones, emails and group chats. Finding a note or an assignment brief at the last minute is stressful, and deadlines get missed. Our project is a File Management AI Assistant. It keeps school files in one place, helps students find and download them, and reminds them when a deadline is close.

This report shows how we applied Human-Centered Design (ISO, 2019) to our choices of colour, fonts, wording and layout, so that the system works for all students and not only for some (Norman, 2013).

2. The Project and Its Users

Users: students of our school. They use different devices (phones, laptops, lab computers), have different internet speeds, and some have low vision or colour blindness. The system is written in English only, so we keep the words short and simple.

Main features:

- One place to store and download course files.
- An AI assistant that helps students find the file they need.
- Notifications that remind students when a deadline is near.

The standard ISO 9241-210 asks designers to start from the user: understand who they are, list their needs, design, then test with real users and improve (ISO, 2019). We follow the same steps.

3. Design Decisions

3.1 Colour

We chose navy blue as the main colour. It looks calm and serious, which suits a school tool, and it gives very strong contrast with white. The other colours are kept few and each one has one job.

Role	Colour	Hex	Contrast	Used for
Main	Navy blue	#0A2A5E	13.9 : 1 on white	Header, buttons, links
Background	White	#FFFFFF	-	Page background
Surface	Light blue-grey	#E8EEF7	12.0 : 1 with navy	Cards, file rows
Reminder	Amber	#F5B301	7.5 : 1 with navy	Deadline alerts (navy text on it)
Error	Red	#B3261E	6.5 : 1 with white	Failed download, wrong input
Success	Green	#1B6E3C	6.3 : 1 with white	File saved, reminder set

- Normal text must reach at least 4.5 : 1 contrast, as required by WCAG 2.2 (W3C, 2023). All colours above pass.
- Colour is never the only signal. A deadline alert also has a bell icon and the word "Due" (W3C, 2023).
- A dark mode (white text on navy) will be offered. Course notes explain that LCD, LED and OLED screens show brightness and black levels differently (University of Rwanda, 2026), so one colour can look weaker on another screen. A choice of modes helps.

3.2 Fonts and Sizes

We use Noto Sans, with Arial as backup. Its letters are clear and open, and it is free to use. Sizes follow the readability rules in WCAG 2.2 (W3C, 2023).

Element	Size	Style
Page title	28 px	Bold
Section heading	22 px	Bold
Body text	16 px (minimum)	Regular, line spacing 1.5
Buttons and labels	16 px	Bold
Help and date text	14 px (smallest allowed)	Regular

- Text is left-aligned, with short lines. No long paragraphs in capital letters.
- Users can enlarge text up to 200% without the page breaking (W3C, 2023).

3.3 Wording

- Short, plain English words. No technical terms without an explanation.
- Error messages say what went wrong and how to fix it, for example "File too large. Choose a file under 20 MB." (Norman, 2013).
- Dates are written the same way everywhere, for example 12/11/2026.

3.4 Sharp on Every Screen

Screens are made of pixels, and resolution and pixel density decide how sharp things look (University of Rwanda, 2026). Students will open the system on very different screens, so we:

- use SVG (vector) icons. Unlike raster images, they stay sharp at any size, the same way vector displays draw clean lines (University of Rwanda, 2026);
- size everything in relative units, so the layout adapts to phone, laptop and lab screens;
- test on a small laptop screen (1366 × 768), a full HD screen (1920 × 1080) and a phone.

3.5 Other Inclusive Choices

- Buttons at least 44 × 44 px so they are easy to tap (W3C, 2023).
- Full keyboard use and screen-reader labels on all buttons and fields.
- The same layout on every page, so users learn it once (Norman, 2013).

4. How the Design Meets User Needs

User need	Our design decision
I use an old phone or a slow connection	Light pages, small file sizes, sharp SVG icons
I find technical words hard	Short, plain English and clear error messages
I have low vision or colour blindness	High contrast, text enlargeable to 200%, icons plus words
I forget deadlines	Reminders by notification and SMS/email, with a snooze button
I do not know the file name	AI assistant finds files from simple questions in plain English
I tap the wrong button	Large buttons, confirmation before delete, undo

5. Conclusion

Human-Centered Design means we design for the students, not for ourselves (ISO, 2019). By choosing navy blue with strong contrast, clear fonts in readable sizes, simple English, and layouts that stay sharp on any screen, our File Management AI Assistant becomes easier to use for every student (W3C, 2023). The next step is to test a prototype with students from our school and improve it from their feedback.

6. References

[1] International Organization for Standardization. (2019). ISO 9241-210:2019 Ergonomics of human-system interaction - Part 210: Human-centred design for interactive systems. ISO.

[2] Norman, D. A. (2013). The design of everyday things (Revised and expanded ed.). Basic Books.

[3] University of Rwanda. (2026). Unit 2: Graphics Display Systems [Course slides]. Department of Computer Science.

[4] World Wide Web Consortium. (2023). Web Content Accessibility Guidelines (WCAG) 2.2. W3C.